Site Name		
Head End & Consoles (Direct runs only - no daisy chaining)		
Device	**Number Per System**	**Power Supplies Required**
R5KM8PRT - Ethernet Switch		0
R5KMSC - Main System Controller		0
351010 - Responder Network Concentrator		0
351006 - Fiber Optic Adapter Module		0
R5KCONS - VOIP Console		
R5KL2KA - L-Net to K-Bus Adapter		
R5KMPR36 Power Supplies Required		0

Notes:

It is assumed each R5KCONS is powered either via the R5KMSC or the R5KM8PRT.

It is assumed the uplink ports of the R5KM8PRT are only used to interconnect other R5KM8PRT switches.

It is assumed that each R5KMSC will have an unused POE port for configuring with no downtime.

The R5KL2KA is powered by the L-NET via the R5KMSC. The R5KMSC only supplies power to 4 R5KL2KA units max.

R5KMSC current is calculated with the above assumtions (doesn't match labeling). (2 - .328 + 4 x .130 = 2.192A)

If the number of R5KCONS per system turns red, more R5KM8PRTs or R5KMSCs are needed.

If the number of R5KL2KA per system turns red, more R5KMSCs are needed.

It is assumed 351006 is powered directly from a power supply, disregard the required supplies if powered via POE.

4000 Power Calculation Worksheet

Power Supply #1

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2	Power Cable Gauge	Voltage at Node (V)	Will Node Work
	↓Fill in the data in the yellow cells↓				15.5	
				CAT5 (24 AWG, 2pr)		
				CAT5 (24 AWG, 2pr)		
				CAT5 (24 AWG, 2pr)		
				CAT5 (24 AWG, 2pr)		
				CAT5 (24 AWG, 2pr)		
				CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524

← 4.65A MAX →		
2.5A Max	2.5A MAX	2.5A MAX
Run 1	Run 2	Run 3
(mA)	(mA)	(mA)
0.00	0.00	0.00
OK	OK	OK

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light	Power Cable Gauge	Voltage at Node (V)	Will Node Work
		↓ Fill in the data in the yellow cells ↓							15.5	
1								CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
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12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		

↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

Loading

40%

Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
*CAT6 (23 AWG, 4pr)	0.03396
12 AWG	0.01063
14 AWG	0.01686
16 AWG	0.02690

*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

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CAT5 (24 AWG, 2pr)		

Site Name	

R4KANNV2 and R4KMQCV2

Run	Device Model	Cable Length (m)	Number of R4KANNV2s Wired in Series With This R4KANNV2
	↓Fill in the data in the yellow cells↓		

Corridor Lights & Domeless Controllers

Run	Node #	Device Model	Cable Length (m)	Number of Enhanced Stations in room	Number of R4KFB1s in room	Number of R4KFAMs in room	Check if CL is a Zone Light
↓ Fill in the data in the yellow cells ↓							
1							

Unit Converter				
	From		To	
Length		feet	0	meters(m)
		yards	0	meters(m)
		meters(m)	0	feet
		meters(m)	0	yards
Current		amps(A)	0	milliamps(mA)
		milliamps(mA)	0	amps(A)

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		
CAT5 (24 AWG, 2pr)		

Cumulative Supply Current	Available Supply Current	Total K-Bus Length Used	Remaining K-Bus Length (Meters)
0.00	4650.00	0	1524
	← 4.65A MAX →		
	2.5A Max	2.5A MAX	2.5A MAX
	Run 1	Run 2	Run 3
	(mA)	(mA)	(mA)
	0.00	0.00	0.00
	OK	OK	OK

Power Cable Gauge	Voltage at Node (V)	Will Node Work
	15.5	
CAT5 (24 AWG, 2pr)		
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↓ Options ↓

- ☐ Staff Registration
- ☐ Audio Paging

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Wire Gauge Resistance	
Standard K-Bus Run	
AWG	ohms/m
CAT5 (24 AWG, 2pr)	0.08419
CAT6 (23 AWG, 2pr)	0.06791
Remote Power Run	
AWG	ohms/m
*CAT5 (24 AWG, 4pr)	0.04209
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12 AWG	0.01063
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*4pr assumes all 4 pairs will be used for power. Remote power runs require an R4KKBSP for powering K-Bus devices.

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CAT5 (24 AWG, 2pr)		

4000 Power Calculation Worksheet Estimate

Device Model		Number Per System	Typ Power Draw (A)	Total Draw Per System		Enter Cost	Subtotal
Stations							
HSS400	High Security Bed Station		0.010				
HSS401	High Security Staff Station		0.008				
HSS433	High Security Push Button Station		0.015				
R4K11V	Single Bed Visual Station		0.002				
R4K12A	Single Bed Audio Station		0.005				
R4K12AHZ	Horizontal Single Bed Audio Station		0.005				
R4K13VA	Single Bed Enhanced Audio Station		0.025				
R4K17V	SLIM Single Bed Enhanced Audio St		0.025				
R4K13VAH	Horiz. Single Bed Enhanced Audio Stn		0.025				
R4K14SA	Staff Assist Bed Station		0.007				
R4K15V	SLIM 1/4" Jack Button Station		0.007				
R4K16LV	Single Bed Enhanced Visual Station		0.015				
R4K21V	Dual Bed Visual Station		0.004				
R4K22A	Dual Bed Audio Station		0.007				
R4K2JACK	SLIM Dual 1/4" Jack Audio Station		0.007				
R4K23VA	Dual Bed Enhanced Audio Station		0.030				
R4KCB10	Code Station		0.007				
R4KCB12	Code Station w/Timer		0.017				
R4KCB13	SLIM Code Station w/Timer		0.017				
R4KCNCL	Cancel Station		0.002				
R4KDLC2	Dual Lighting Controller		0.023				
R4KDY	Duty Station		0.012				
R4KDTY2	SLIM Duty Station		0.012				
R4KESR	Emergency Staff Reg. Station		0.007				
R4KSAR	SLIM Dual Push w-Staff Reg Station		0.007				
R4KFAM	Fire/Auxiliary Module		0.030				
R4KFB1	Feature Bed Interface		0.075				
R4KMST	Marquee Speaker		0.005				
R4KPB11	Push For Help Station		0.004				
R4KPB12	SLIM Push For Help Station		0.004				
R4KPB22	Dual Push Button Station		0.017				
R4KPB23	SLIM Dual Push Button Station		0.017				
R4KPB44	Four Button Station		0.014				
R4KPC10	Pull Cord Station		0.002				
R4KPC11	SLIM Pull Cord Station		0.002				
R4KRA1	Residence Assist Station		0.007				
R4KSLC1	Single Lighting Controller		0.013				
R4KSR1	Staff Registration Station		0.002				
R4KSS	Staff Station		0.005				
R4KTVR1	Digital TV Isolation Module		0.015				
R5KPS1V	Responder 5k Single Pat Visual		0.003				
R5KPB4	Responder 5k Four Button St		0.011				
R5KPD2EA	Responder 5k Enhanced Dual Audio Patient Station		0.015				
R5KPS1EA	Responder 5k Enhanced Single Audio Patient Station		0.011				
R5KPD2A	Responder 5k Dual Audio Patient Station		0.010				
R5KPS1A	Responder 5k Single Audio Patient Station		0.008				
R5KSSTF	Responder 5k Staff Station		0.016				
R5KSDTY	Responder 5k Duty Station		0.027				
R5KPS1LCA	Responder 5k Single Audio Patient Station with Button and Light Control		0.031				
Corridor Light/Domeless Controllers							
DCV100	6 Point Visual Domeless Controller		0.012		Loading		
R4KOUT4R	Visual Output Controller		0.032				
DCV116	16 Point Domeless Controller		0.012				
CLV122	2 Bulb Visual Corridor Light		0.012				
CLV144	4 Bulb Visual Corridor Light		0.012				
DCA200	6 Point Audio Domeless Controller		0.038				
R4KOUT4S	Audio Output Controller		0.021				
DCA216	6 Point Audio Domeless Controller		0.038				
CLA214D	Duty Corridor Light		0.058		60%		
DCA214D	Duty Domeless Controller		0.058				
CLA222	2 Bulb Audio Corridor Light		0.038				
CLA244	4 Bulb Audio Corridor Light		0.038				
CLAR4	Corridor Light Audio Relay Add-on		0.032				
CLA246	4 Bulb Audio Corridor Light		0.038				
CLAR46	Corridor Light Audio Relay Add-on		0.032				
R5KCL546	CL 4 audio 6 stations		0.058				
R5KDC46	Domeless 4 audio 6 stations		0.038				
R5KCL516	CL 1 audio 6 stations		0.038				
R5KCL506	CL 6 stations		0.036				
R5KDC06	Domeless 6 stations		0.016				
R5KDC016	Domeless 16 stations		0.017				
R5KDC16D	Domeless Duty 1 Audio 6 stations		0.058				
R5KDCRC4	Relay Output Controller		0.032				
R5KDCRC4	Solid State Relay Output Controller		0.021				
Consoles and Annunciators							
R4KANNV2	Annunciator Panel		0.375				
R4KMQCV2	Marquee Controller		0.300				
R5KCONS	VOIP Console		0.328				
Network and Control Equipment							
R4KPA25	25W Paging Amp		2.500				
R5KMSC	Main System Controller		2.192				
R5KL2KA	L-Net to K-Bus Adapter		0.130				
351010	Responder Network Concentrator		0.300				
R5KM8PRT	R5K 8 Port POE Ethernet Switch		2.400				
351006	Fiber Optic Adapter Module		0.075				
Device Model		**Number Per System**	**Power Draw (A)**	**Total Draw Per System**			
Stations							
			Grand Total (15V)	0.000			
			Grand Total (36V)	0.000			
R4KPR400 Power Supplies Required				**0**			
R5KMPR15 Power Supplies Required				**0**			
R5KMPR36 Power Supplies Required				**0**			
						Total	$0.00

Notes:
It is assumed the R5KCONS is powered either via the R5KMSC or the R5KM8PRT.
It is assumed the uplink ports of the R5KM8PRT are only used to interconnect other R5KM8PRT switches.
It is assumed that each R5KMSC will have an unused POE port for configuring with no downtime.
The R5KL2KA is powered by the L-NET via the R5KMSC. The R5KMSC only supplies power to 4 R5KL2KA units max.
R5KMSC current is calculated with the above assumtions (doesn't match labeling). (2 - .328 + 4 x .130 = 2.192A)
If the number of R5KCONS per system turns red, more R5KM8PRTs or R5KMSCs are needed.
If the number of R5KL2KA per system turns red, more R5KMSCs are needed.
It is assumed 351006 is powered directly from a power supply, disregard the required supplies if powered via POE.

Responder 5000 Power Calculation Workbook

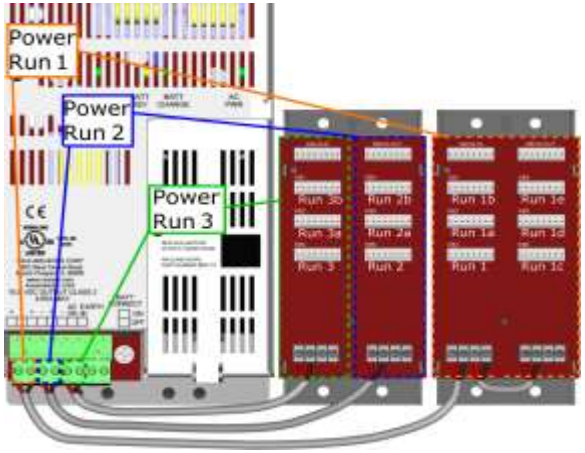
For further notes please refer to the manual
KI-2907 - Responder 5000 Installation Guide

This workbook should be used to assist you in planning the wiring of a Responder 5000 installation.

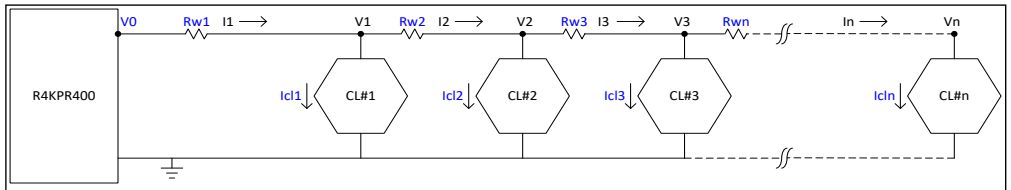
Each worksheet (yellow tabs 1-20) helps you to determine the number of devices that can be powered by a single R5KMPR15 Power Supply based on device selection, cable length, and layout variables. Please keep in mind that the entire workbook including all tabs accounts for a complete Responder 5000 system with multiple R5KMSCs, R5KL2Ks and multiple R5KMPR15s. Certain, very large, configurations will require multiple instances of the workbook to account for all devices of the Nurse Call System.

In addition, the utility provides a "pre-sale" power supply estimate worksheet. This can be used to help you estimate the number of power supplies a site will need. The calculations are conservative to reserve room for additions in the future when detailed configurations are completed.

The spreadsheets assume runs 1, 2, and 3 are corridor lights runs wired in series from three (3) separate R5KMTRM runs. For parallel runs (when one (1) output of the supply powers two (2) R5KMTRMs runs), select 1-1e, 2-2e or 3-3e. See above image for sample runs. These runs are calculated as if they were parallel runs on an R5KMTRM. R4KANNV2s and R4KMQCV2s are tracked separately from CLs but included on the same power supply. Attention should be paid to the total current display for each run to stay within the power supply unit limits.



Single power supply example



Icln = In
Vn = V3 - (In x Rwn)

Vn = V3 - (Icln x Rwn)

I3 = In + Icl3
V3 = Icl3 x Rcl#3
V3 = V2 - (I3 x Rw3)

V3 = V2 - [(In + Icl3) x Rw3]
V3 = V2 - [(Icln + Icl3) x Rw3]

I2 = I3 + Icl2
V2 = Icl2 x Rcl#2
V2 = V1 - (I2 x Rw2)

I2 = In + Icl3 + Icl2
I2 = Icln + Icl3 + Icl2
V2 = V1 - [(Icln + Icl3 + Icl2) x Rw2]

I1 = I2 + Icl1
V1 = Icl1 x Rcl#1
V1 = V0 - (I1 x Rw1)

I1 = In + Icl3 + Icl2 + Icl1
I1 = Icln + Icl3 + Icl2 + Icl1
V1 = V0 - [(Icln + Icl3 + Icl2 + Icl1) x Rw1]